

Jason Hong

626-560-6545 | jasonph@mit.edu | [linkedin.com/in/jasonphong](https://www.linkedin.com/in/jasonphong) | github.com/jasonph314

EDUCATION

Massachusetts Institute of Technology

Cambridge, MA

B.S. in Computer Science and Engineering

Sept 2025 - May 2028

- GPA: 5.0/5.0
- Relevant Coursework: Computation Structures, Algorithms, Computational Thinking and Data Science, Fundamentals of Programming
- Activities and societies: Chinese Students and Scholars Association

EXPERIENCE

Durability

Cambridge, MA

Incoming Software Engineer Intern

June 2026 - Aug 2026

- Built a centralized **SwiftUI state machine** to manage complex 7-movement assessment flows.
- Integrated **Asensei SDK** for on-device pose estimation and clinical-grade athletic feedback.
- Developed a 6-layer scoring pipeline to normalize raw CV data into body-part performance metrics.
- Managed **Supabase RLS** configurations to ensure secure data isolation and multi-tenant privacy.

Kyte Dynamics

Los Angeles, CA

Software Engineer Intern

May 2025 - Sept 2025

- Developed embedded C/C++ PX4 drivers using **MAVLink** for low-latency telemetry synchronization.
- Refactored **2,000+** lines of legacy C/C++ into modular abstractions to reduce technical debt.
- Optimized CI/CD pipelines via CMake parameters, **reducing compile times by 50%**.
- Built MATLAB/Simulink test benches to validate IMU performance and analyze hardware failure modes.

RESEARCH

MIT Laboratory for Information & Decision Systems

Cambridge, MA

Undergraduate Researcher (UROP) | Advised by Prof. Tamara Broderick

April 2026 - Present

- Integrating **TextSlinger API** into the **Nomon** simulation framework to support SOTA language models.
- Benchmarking LLM word-prediction performance on selection speed for single-switch interfaces.
- Refactoring probabilistic behavior models to align with modern NLP-driven inference workflows.

HONORS & AWARDS

MIT Pozen Fellowship: Awarded to select students for excellence in engineering and public service (2026)

Dr. Sherrie Emoto Mathematics Scholarship: Highest departmental honor for mathematical achievement at Fullerton College (2025)

PROJECTS

Open Source Contributions | *Lua, Neovim API, Async I/O*

Nov 2025 - Present

- Engineered a **Cursor-style inline editor** for **claude-inline.nvim** using the Neovim API to enable direct buffer prompting.
- Resolved critical **async race conditions** in Lua to ensure seamless cursor control post-AI code generation.
- Extended **opencode.nvim** backend architecture to support **local open-source LLMs**, increasing user privacy and flexibility.

RISC-V Neural Network Inference Accelerator | *Minispec/BSV, C, RISC-V*

Spring 2026

- Optimized a pipelined processor and MNIST inference stack to placing **top 10; 80×** improvement over baseline.
- Extended the ISA with custom instructions and a pipelined SIMD multiplication to reduce clock period by **51%**
- Reduced clock period **572ps** → **529ps** by re-encoding ALU branch output as a single comparison bit; implemented a Kogge-Stone adder, specialized iCache, and branch target buffer (BTB) to minimize CPI.

TECHNICAL SKILLS

Languages: C/C++, Python, Swift, Minispec, Bash

Tools & Frameworks: Linux/Unix, Git, Neovim, SwiftUI, Supabase, CMake, MAVLink

Interests: Computer Architecture, Verified Software Engineering, OS Kernels, Performance Engineering